MILITARY SPECIFICATION

DIPHENHYDRAMINE HYDROCHLORIDE SYRUP

This specification is approved for use by the Defense Personnel Support Center, Defense Logistics Agency, and is available, for all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Diphenhydramine Hydrochloride Syrup (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-C-186 - Containers, Packaging and Packing for Drugs, Chemicals and Pharmaceuticals.

STANDARDS

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 S. 20th St., Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 6505

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE, FOOD AND DRUG ADMINISTRATION (FDA)

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia (USP).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

The National Formulary (NF).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material. Shall be a pleasant tasting syrup containing in each 5 ml:

Diphenhydramine Hydrochloride	12.5 mg
Ammonium Chloride	125.0 mg
Sodium Citrate	50.0 mg
Glycerin	390.62mg
Menthol	1.0 mg
Alcohol	5% by volume

3.2 Additives. The coloring, flavoring and sweetening agents and preservatives used in the syrup shall be in the amounts approved by the FDA.

3.3 Assay of the finished syrup.

3.3.1 Diphenhydramine hydrochloride. Shall contain not less than 12.00 mg and not more than 13.25 mg of diphenhydramine hydrochloride in each 5 ml, when assayed as specified in 4.6.1.

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3.3.2 Alcohol. Shall contain not less than 4.0 percent and not more than 6.0 percent by volume of alcohol, when assayed as specified in 4.6.2.

3.4 Ammonium chloride, sodium citrate, and menthol. The quantities of ammonium chloride, sodium citrate, and menthol shall be verified by the cognizant government inspector from the batch production records.

3.5 Palatability. The syrup shall be palatable and pleasant to the taste with no unpleasant after taste, when chewed. Not later than the time specified for opening of bids or receipt of proposals, the offeror shall submit to the contracting officer six 4 fl oz samples representative of the product which the offeror proposes to furnish. Samples will be subjected to panel testing for a determination of palatability (see 4.6.3 Palatability test). A portion will be used by the cognizant Government inspection and quality assurance activities for determining compliance of supplies furnished hereunder with the palatability requirement. Approval as to palatability of any sample submitted by the offeror will not constitute approval of the sample as to any other requirement of this specification. The requirement for submission of samples for use in determining compliance with the palatability requirement may be waived, provided the offeror states, in his bid or proposal, that the product he proposes to furnish is the same product he has offered to the purchasing activity on a previous procurement and the contracting officer determines that such product was previously procured and/or tested by the purchasing activity and found to comply with the specification requirement.

3.6 Ingredients.

3.6.1 Diphenhydramine hydrochloride, ammonium chloride, sodium citrate, glycerin, menthol and alcohol. The diphenhydramine hydrochloride, ammonium chloride, sodium citrate, glycerin, menthol and alcohol used in the manufacture of syrup shall comply with the tests, standards, and requirements of the USP, including any supplements or revisions thereto.

3.7 Other ingredients. All other ingredients used in the manufacture of the syrup shall be of USP or NF quality, or, if not included in either of these compendia, they shall be suitable for use in this cough syrup.

3.8 Expiration dating period. The expiration dating period shall be not less than 36 months.

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3.8.1 Delivery. Not more than 3 months of the expiration dating period shall have elapsed at the time of delivery to the Government.

3.9 Workmanship. The material and its containers shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item, for the expiration dating period specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the item to the Government.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Lot. For purposes of this specification, a lot, batch, or control is that single, uniform, and homogeneous quantity of syrup produced from one formulation, subjected to the same compounding and manufacturing operation, and filled into final containers.

4.3 Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

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TABLE I. Sampling.

For visual examination	Inspection level	AQL (percent defective	Unit of product
Major	S-2	1.0	Filled immediate container
Minor	II	2.5	Filled immediate container

4.3.1 Raw material and end item testing. Noncompliance with any of the following is cause for rejection.

TABLE II. Criteria for raw material and end item testing.

Characteristic	Requirement paragraph	Inspection level 1/	Minimum sample
Diphenhydramine Hydrochloride Assay	3.3.1	S-2	5 } Composite
Alcohol Assay	3.3.2	S-2	5 } Composite
Palatability Test	3.5	-	-

1/ Where possible, the sample shall be used for two or more tests.

4.4 Examination. The finished syrup shall be examined to determine compliance with all requirements of this specification. Nonconformance will be permitted to the extent indicated in 4.3 and 4.3.1.

4.5 Classification of defects. Examination will be conducted in accordance with the following classification of defects.

TABLE III. Classification of defects.

Categories	Defects 1/
Major	
101	Syrup not free from foreign matter.
102	Syrup has crystal formation.
103	Syrup not homogeneous.
Minor	
201	Exterior of immediate container is not free from contents or foreign matter.
202	Label not free from defacing marks and tears.

1/ Inspection is not restricted to the classified possible defects listed above.

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4.6 Tests.

4.6.1 Assay for diphenhydramine hydrochloride.

Phosphate buffer solution:

Dissolve 34.84 grams of potassium phosphate dibasic (K_2HPO_4) in about 1 liter of purified water in a 2 liter volumetric flask. Add 10 ml of 0.1N hydrochloric acid and dilute to about 1950 ml. Check the pH by the USP potentiometric method and adjust the pH to 8.80 to 8.90 with 0.1N hydrochloric acid acid or 0.1N NaOH, if necessary. Dilute to volume with purified water.

Procedure:

Transfer 10 ml of syrup, by means of a "to contain" pipet into a 25 ml volumetric flask. Let the pipet drain for about 15 minutes then wash it out completely with the phosphate buffer collecting the washings in the volumetric flask. Dilute to volume with the buffer and mix. Transfer with a "to contain" pipet 10 ml of the buffered mixture into a 250 ml beaker containing 13 grams of Celite 545. Let the pipet drain for at least 10 minutes then rinse carefully with 2 to 3 ml of buffer. Mix thoroughly until a uniform mass is obtained. Pack the mass in several (10 to 12) portions into a chromatographic tube (2.5 cm inside diameter by 20 cm length) which was previously packed over a pledget of cotton with a uniform mixture of 5 grams Celite 545 and 4 ml of phosphate buffer. Wipe the rod and the beaker used for the sample with one or two small pieces of cotton slightly moistened with buffer. Add the cotton to the column and press it tightly over the sample. Elute the diphenhydramine base with a mixture of 160 ml n-heptane and 22 ml of $CHCl_3$ (reagent grade) at a rate of 4 to 6 ml per minute. Collect the eluate in a 250 ml separatory funnel. Extract the diphenhydramine base by carefully shaking the eluate with 3 portions (15, 15, 10 ml) of 0.1N H_2SO_4. Collect the acid extracts in a 50 ml volumetric flask, dilute to volume with 0.1N H_2SO_4 and mix carefully (not too vigorously). Prepare a blank by carefully shaking 50 ml 0.1N H_2SO_4 with a mixture of 160 ml n-heptane and 22 ml $CHCl_3$. Filter the sample solution and the blank through a Whatman glass fibre GFC filter paper rejecting the first 5 to 10 ml. Determine the absorbances of the filtered solution and a concomitant Diphenhydramine Hydrochloride USP Reference Standard containing approximately 20 mg per ml in 0.1N H_2SO_4, against an 0.1N H_2SO_4 blank, at 258 nm in a 1 cm cell using a suitable spectrophotometer.

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Calculation:

$$\frac{Au}{As} \times C_s \times 12.5^{1/} \times \frac{2/}{5} = \text{mg diphenhydramine hydrochloride per 5 ml}$$

Where: Au is the absorbance of the sample at 258 nm
As is the absorbance of the standard at 258 nm
Cs is the concentration of the standard in mg per ml

1/ 12.5 is the dilution factor
2/ 5 represents the assay requirement per 5 ml

4.6.2 Alcohol. Transfer 50 ml of finished syrup, accurately measured, and 50 ml of purified water to a distilling flask. Use a portion of the purified water to rinse out the sample measuring flask. Add 2 ml of concentrated Phosphoric Acid (H_3PO_4) and a few porous granules to prevent bumping. Distill and collect about 47 ml of distillate into a 50 ml volumetric flask. Cool the temperature of the distillate to 25°C, dilute to volume with purified water and mix. Transfer quantitatively to a 250 ml separatory funnel and extract with 40 ml of Benzin Purified (Hexane) shaking gently for 3 to 5 minutes. Allow to stand until a clean separation occurs. Drain off about 30 ml of the lower phase and determine the specific gravity at 25°C. Using this result and an alcoholometric table ascertain the percentage, by volume, at 15.56°C (60°F) of alcohol contained in the sample tested.

4.6.3 Palatability test. A taste panel consisting of 10 members will be used to determine acceptability of samples. Samples will be prepared for testing (samples will be tested undiluted), coded, and served to panel members under controlled serving conditions, e.g., all samples will be of the same amount, and served at the same temperature; each panel member will receive an equal number of samples; the order of serving will be varied among panel members; an interval of at least five (5) minutes will elapse between successive samples and panel members will rinse their mouths with water (room temperature) after each sample; panel members will test without interference either from each other or from outsiders. The product offered shall be rated equal to or better than the item's Palatability Standard 1/, when determined by the taste panel, using the following 9-point hedonic rating scale. The average rating of the samples shall be equal to or greater than the average rating of the standard, similarly prepared and tested.

1	2	3	4	5	6	7	8	9
Dislike extremely	Dislike very much	Dislike moderately	Dislike slightly	Neither like nor dislike	Like slightly	Like moderately	Like very much	Like extremely

1/ The item's Palatability Standard is available, upon separate request to the contracting officer, Defense Personnel Support Center, Directorate of Medical Materiel.

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4.7 Packaging inspection. The inspection of the intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.8 Labeling inspection. Sampling shall be in accordance with MIL-STD-105 and the unit of product shall be the unit of issue. The inspection lot shall be the number of units of issue of each batch, lot, control number, etc., offered for inspection at one time. The inspection level for labeling shall be S-2 and the acceptance number shall be zero.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with all applicable requirements of PPP-C-186 and as specified herein.

5.2 Packaging. Packaging shall be level A or C, as specified (See 6.1).

5.2.1 Unit of issue. One bottle containing 4 fl oz, as specified, constitutes one unit of issue.

5.2.2 Level A.

5.2.2.1 Immediate container. The immediate container shall be a commercially available bottle with childproof closure considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatability that neither the contents nor the container are altered in any way by each other.

5.2.2.2 Unit package. At the option of the contractor, each unit may be packaged as specified in 5.2.5 of PPP-C-186.

5.2.2.3 Intermediate package. Twelve units of issue shall be packaged as specified in 5.2.8 of PPP-C-186.

5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Packing. Packing shall be level A, B or C, as specified (see 6.1).

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5.3.1 Level A. Forty-eight (48) units of issue shall be packed as specified in 5.4.4 of PPP-C-186.

5.3.2 Level B. Forty-eight (48) units of issue shall be packed as specified in 5.4.3.2 of PPP-C-186.

5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act and shall include the information required below:

5.4.1.1 Immediate container. Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.

(b) the expiration date.

5.4.1.2 Unit package . Each unit package label shall bear the following information:

(a) label information in accordance with commercial practice.

(b) the National Stock Number.

NOTES: (1) The National Stock Number shall not be preceded by any prefix.
(2) Strip labeling may be used.
(3) Label adhesion test shall not apply.
(4) If a unit package is not supplied, the National Stock Number shall appear on the immediate container label.

5.4.2 Intermediate package. Intermediate package shall be marked as specified in 5.5.3 of PPP-C-186. Type I Shelf Life marking as specified in MIL-STD-129 shall be shown. Marking shall include the legend: "STORE AT CONTROLLED ROOM TEMPERATURE 15° - 30°C (59° - 86°F)".

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5.4.3 Exterior container. Exterior container shall be marked as specified in 5.5.4 of PPP-C-186. Type I shelf life marking as specified in MIL-STD-129 shall be shown. Marking shall include the legend: "STORE AT CONTROLLED ROOM TEMPERATURE 15°-30°C (59°-86°F)".

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) National Stock Number.
(c) Selection of applicable levels of packaging and packing (see 5.2 and 5.3).

6.2 This specification covers the following item(s):

National Stock No.	Item Identification
6505-01-029-8074	DIPHENHYDRAMINE HYDROCHLORIDE SYRUP, 4 fl oz, (118 ml)

Preparing activity:
DLA-DM
Project No. 6505-1870

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE
MIL-D-37727 (DLA-DM) DIPHENHYDRAMINE HYDROCHLORIDE SYRUP

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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